

Shreya Nakum

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 github.com/sn82978

 sn82978.github.io/

Education

University of California, Irvine

Expected June 2027

Bachelor of Science in Computer Science with Honors (GPA: 3.80 / 4.00)

Irvine, CA

- **Relevant Coursework:** Programming in C/C++, Data Structures & Algorithms, Discrete Math, Operating Systems, Applications of Probability in Computer Science, Machine Learning & Data Mining.
- **Honors/Awards:** 1st Place Best Collaborative Research at UCI UROP 2025, Chancellor's Award for Excellence in Undergraduate Research, HackMIT Modal Honorable Mention, 6x Dean's Honors List.
- **Involvements:** Information and Computer Science Honors Program, Campus Honors Collegium, Artificial Intelligence@UCI

Technical Skills

Languages: Python, C++, C, R, Javascript/Typescript, Java, SQL, MIPS (Assembly), MATLAB, Fortran 77, Fortran 90.

Frameworks & Libraries: PyTorch, TensorFlow/Keras, Flask, FastAPI, BeautifulSoup, scikit-learn, transformers, Robot Operating System (ROS).

Tools & Platforms: AWS (Lambda, Boto3), Firebase, Docker, PostgreSQL, Milvus, HuggingFace, Git, WordPress, Simulink, Linux.

Expertise: Computer vision, CNNs, RAG, Data Analysis, Statistical Modeling, Physical Simulations, Academic Writing.

Other: Web Development, API Integration, Web Scraping, MLOps, Supercomputing.

Experience

NASA: Johnson Space Center

Jan 2026 – Aug 2026

Guidance, Navigation, and Control Intern

Houston, Texas

- Implemented high-fidelity **attitude dynamics simulations** for LEO commercial space stations with **Fortran 77, C,** and **MATLAB/Simulink**.
- Implemented **reaction wheel actuator models** and integrated them into a closed-loop **attitude control systems**.
- Developed computer vision program to **detect lasers independent of color for altitude during terminal descent** of Artemis IV lander module to connect to continuous lasers & control a frame camera using ROS.

National Science Foundation: University of Oulu - M3S Lab

June 2025 – Aug 2025

Researcher

Oulu, North Ostrobothnia, Finland

- Built a RAG pipeline to measure code clone similarity in software repositories.
- **Benchmarked 3 LLMs** across zero-shot, few-shot, and chain-of-thought prompts with 3k+ thresholding values.
- Automated statistical evaluation of model metrics; co-authoring a research paper.

UC Irvine - Rose Labs (School of Biological Sciences)

Sep 2024 – Present

Researcher

Irvine, CA

- Trained **220 supervised ML models** to identify evolving DNA sequences, achieving **80% accuracy**.
- Designed a **CNN object detection pipeline (95% accuracy)** for small-scale biological imaging.
- Developed a **multi-camera tracking algorithm** implementing **3D pose estimation and identity propagation** for dense biological assays.
- Deployed a **Flask+JS** classification system that **accelerated model inference by 10x**.
- Awarded 2 research prizes for novel small-object detection methods; currently drafting publication.

dabbl

June 2024 – Aug 2024

Software Engineer Intern

Los Gatos, CA

- Scraped Common Data Set reports across 6k+ colleges using **BeautifulSoup, requests, and pdfplumber**.
- Shipped an in-app **Firebase** notification system for 500+ active users.

Atoma Media

Nov 2023 – Nov 2024

Large Language Model Specialist

Irvine, CA

- Released a HuggingFace sentiment analysis model (**80% accuracy, 6k+ downloads**)
- Connected HuggingFace API to production frontend via TypeScript.
- Built and automated Twitter scraper with PostgreSQL + AWS Lambda.

California Institute of Technology

Oct 2021 – Mar 2023

Researcher

Pasadena, CA

- Processed Hubble COS-Halos datasets with Python (**astropy, numpy, pandas**)
- Derived galaxy halo mass and velocity profiles; visualized results with **matplotlib**
- Published findings in *MNRAS* and presented it at the American Astronomical Society conference

Projects

HackMIT: ActsAI | Honorable Mention, Modal.AI

[GitHub](#)

- Built an Llama-based chatbot mental health platform connecting users to therapists; integrated SMS alerts for crisis detection.

Artificial Intelligence@UCI: Workshop Coordinator

[GitHub](#)

- Give lectures on Machine Learning concepts with hundreds of views.

Publications

Butsky, I., Nakum, S., Ponnada, S., Hummels, C., Ji, S., & Hopkins, P. (2022). Constraining Cosmic-ray Transport with Observations of the Circumgalactic Medium. *Monthly Notices of the Royal Astronomical Society*, **521**(2), 2477–2483. <https://doi.org/10.1093/mnras/stad671>

Nakum, S., Kemman, J., Esposito, M., Cerny, T., Lenarduzzi, V., & Taibi, D. (2025). Leveraging LLMs for Code Clone Similarity Classification in Open Source Microservice Repositories. *Automated Software Engineering*, **[In-Review]**

Nakum, S., & Qi, M. (2025). Drift and Anti-Drift: Hundreds of Generations of Evolutionary Stasis. Presented at the UC Irvine Undergraduate Research Symposium, Irvine, California